



BITS Pilani
Pilani | Dubai | Goa | Hyderabad

Prayogshala GPU Lab Manual

CSIS Dept, WILP

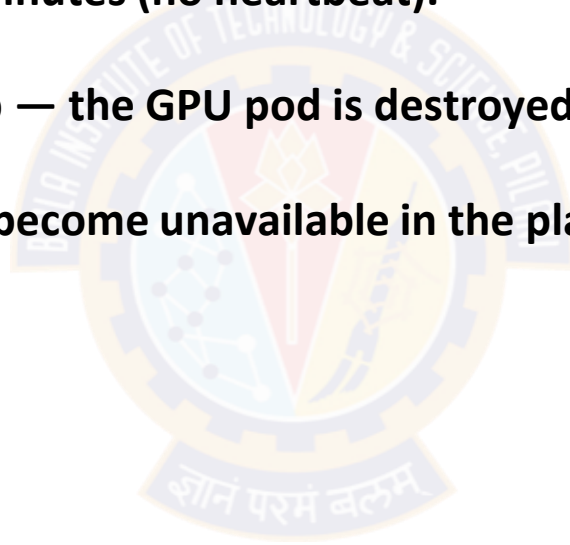
Lab Portal URL

<https://prayogshala.bits-pilani.ac.in>



Important instructions

- 💰 GPU usage consumes the allocated lab budget while the session is running.
- 🕒 Sessions auto-terminate if idle for 15 minutes (no heartbeat).
- 🛑 Save your work before stopping the lab — the GPU pod is destroyed on stop.
- 🛑 At times, GPU might be unavailable or become unavailable in the platform during the launch. Please try launching the lab after a few minutes.



Step 1 - Open Prayogshala Portal

Click 'Continue with BITS SSO'.

The screenshot shows a web browser window with the address bar displaying `https://stg.bitslabs.in/auth/login`. The page features the BITS logo and the text "WORK INTEGRATED LEARNING PROGRAMMES" above the "Prayogshala" heading. Below the heading, it states: "Hands-on engineering, without the campus. Work directly on high-end, industry-grade infrastructure — GPU clusters, cloud machines and real lab hardware — provisioned for your programme and ready in your browser, wherever you work."

Under the heading "First time here?", it says "Lab access is provisioned by your department coordinator." and a link for "Contact IT".

There are five lab categories listed in boxes:

- Virtual Labs**: Cloud VMs
- Cluster Labs**: GPU / CPU Containers
- GPU Labs**: On-demand GPUs
- Remote Labs**: Physical Hardware
- Console Labs**: Cloud Sandbox

On the right side, a "Sign in to BITS Prayogshala" box contains a blue button labeled "Continue with BITS SSO →". A red arrow with the text "Click here" points to this button. Below the button, a note states: "You'll be redirected to the BITS Pilani identity provider, then returned here."

At the bottom of the page, there are links for "Help Center", "Privacy Policy", and "Contact IT", followed by a disclaimer: "An institution deemed to be a University estd. vide Sec.3 of the UGC Act, 1956."

The Windows taskbar at the bottom shows the system clock as 11:47 on 30-07-2026, with weather information (24°C, Light rain) and various application icons.

Step 2 - Login with BITS Credentials

Enter your BITS email ID and password of eLearn portal and click 'Log In'.

Login - eLearn Integrated Portal

https://idp.bits-pilani.ac.in/idp/Authn/UserPassword

 **WORK
INTEGRATED
LEARNING
PROGRAMMES**

BITS Email Address

cloud@wilp.bits-pilani.ac.in

Password

.....

☐ Remember Me [Forgot password?](#)

Click here → **Log In >**

[Click here for login instructions](#)

An institution deemed to be a University estd. vide Sec.3 of the UGC Act, 1956
under notification #F.12-23/63.U-2 of June 18, 1964

innovate achieve lead

24°C Light rain Search 11:48 30-07-2026

Step 3 - Launch GPU Lab

Select your GPU Lab course and click 'Launch Lab'.

The screenshot shows the Prayogshala Lab Management Portal dashboard. The browser address bar displays <https://prayogshala.bits-pilani.ac.in/dashboard>. The page header includes the BITS Pilani logo, the Prayogshala Lab Management Portal title, a user dropdown menu set to 'Student', a notification bell with a red '7', and a 'Logout' link. The left sidebar contains a 'Dashboard' button. The main content area features two course cards. The first card, 'AIMLG549 API DRIVEN CLOUD NATIVE SOLUTIONS', is marked as 'Virtual' and includes tags for 'API Gateway', 'AWS Billing and Cost Services', and 'Bedrock'. The second card, 'CSIST112 DUMMY GPU', is marked as 'GPU' and includes a 'Jupyter Notebook' tag. Both cards have a 'Launch Lab' button. A red arrow points to the 'Launch Lab' button of the 'DUMMY GPU' course, with the text 'Click here' overlaid on the arrow. The bottom of the page shows a 'Csis Student STUDENT' profile card and a database icon.

Step 4 - Select a GPU

Select an available GPU and click 'Launch GPU Lab'.

The screenshot shows a web browser window with the URL <https://prayogshala.bits-pilani.ac.in/dashboard>. The page displays the 'GPU Lab Session' interface for 'DUMMY GPU - CSIST112'. A yellow warning box states: 'Sessions auto-terminate if idle for 15 minutes (no heartbeat)'. Below this, two boxes show 'BUDGET REMAINING \$1.93 / \$2.00' and 'GPU TYPE NVIDIA A40'. The 'Select a GPU' section lists three options: 'NVIDIA A40' (Available), 'RTX 4090' (Available), and 'NVIDIA L40' (Limited). A red box highlights the 'NVIDIA A40' option. A red arrow points from the text 'Click here' to a blue 'Launch GPU Lab' button. A 'Close' button is visible at the bottom left of the modal.

GPU Lab Session
DUMMY GPU - CSIST112

- Sessions auto-terminate if idle for 15 minutes (no heartbeat)

BUDGET REMAINING
\$1.93 / \$2.00

GPU TYPE
NVIDIA A40

Select a GPU

- ☒ NVIDIA A40 Available
- ☐ RTX 4090 Available
- ☐ NVIDIA L40 Limited

Click here

Launch GPU Lab

Close

Step 5 - GPU Environment Provisioning

Wait while the GPU lab environment is being created (usually 2-3 Minuets).

The screenshot shows a web browser window with the URL <https://prayogshala.bits-pilani.ac.in/dashboard>. The page displays the BITS Lab Management Portal interface. A modal dialog titled "GPU Lab Session" is open, showing the session name "DUMMY GPU - CSIST112". The dialog includes a list of important instructions:

- USD is deducted every minute while the session is running
- Save your work before stopping — the pod is destroyed on stop
- Sessions auto-terminate if idle for 15 minutes (no heartbeat)

Below the instructions, a large circular progress indicator is shown with the text "Creating GPU Environment..." and "This usually takes 30–120 seconds. Please wait." At the bottom of the dialog, there is a "Credit History" link and a status indicator "1.93 / 2.00 remaining". A "Close" button is located at the bottom right of the dialog.

Step 6 - Open the GPU Lab

Once the status changes, click 'Open Lab' to access the JupyterLab environment.

The screenshot displays the BITS Lab Management Portal interface. The main content area shows a 'GPU Lab Session' for 'DUMMY GPU - CSIST112'. The session status is 'RUNNING'. A yellow warning box indicates 'Sessions auto-terminate if idle for 15 minutes (no heartbeat)'. The session is associated with 'A40' hardware. The budget remaining is \$1.93, and the amount spent is \$0.00. The duration is 0 min. A red arrow points to the 'Open Lab' button, which is highlighted with a red box. The 'Stop Lab' button is also visible. The interface includes a sidebar with navigation options like 'Dashboard', 'API DR', and 'API Gateway'. The bottom of the interface shows a 'Credit History' dropdown and a 'Close' button.

Click here

GPU Lab Session
DUMMY GPU - CSIST112

- Sessions auto-terminate if idle for 15 minutes (no heartbeat)

• RUNNING

A40

BUDGET REMAINING
\$1.93

SPENT
\$0.00

DURATION
0 min

Open Lab

Stop Lab

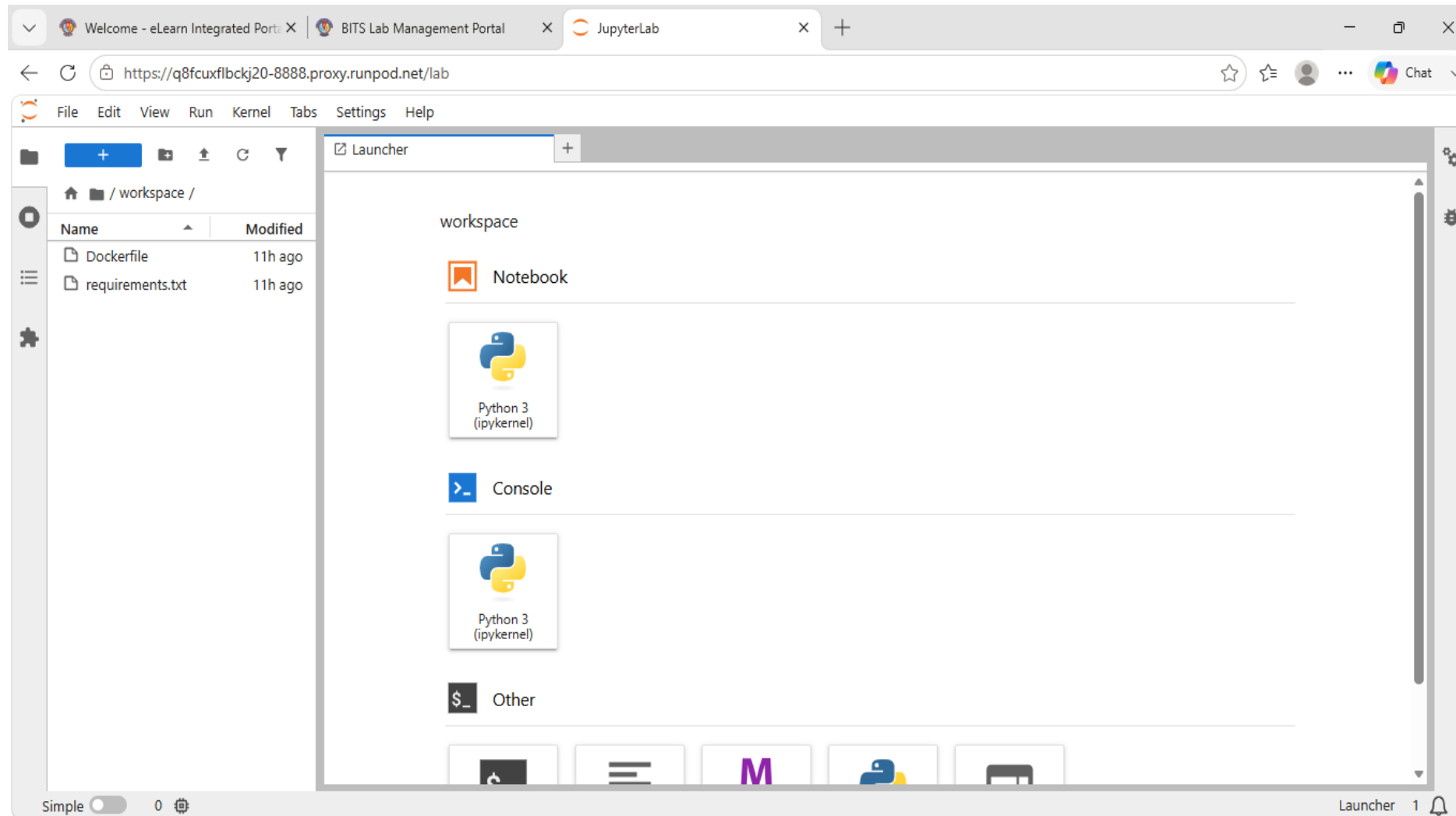
Credit History

1.93 / 2.00 remaining

Close

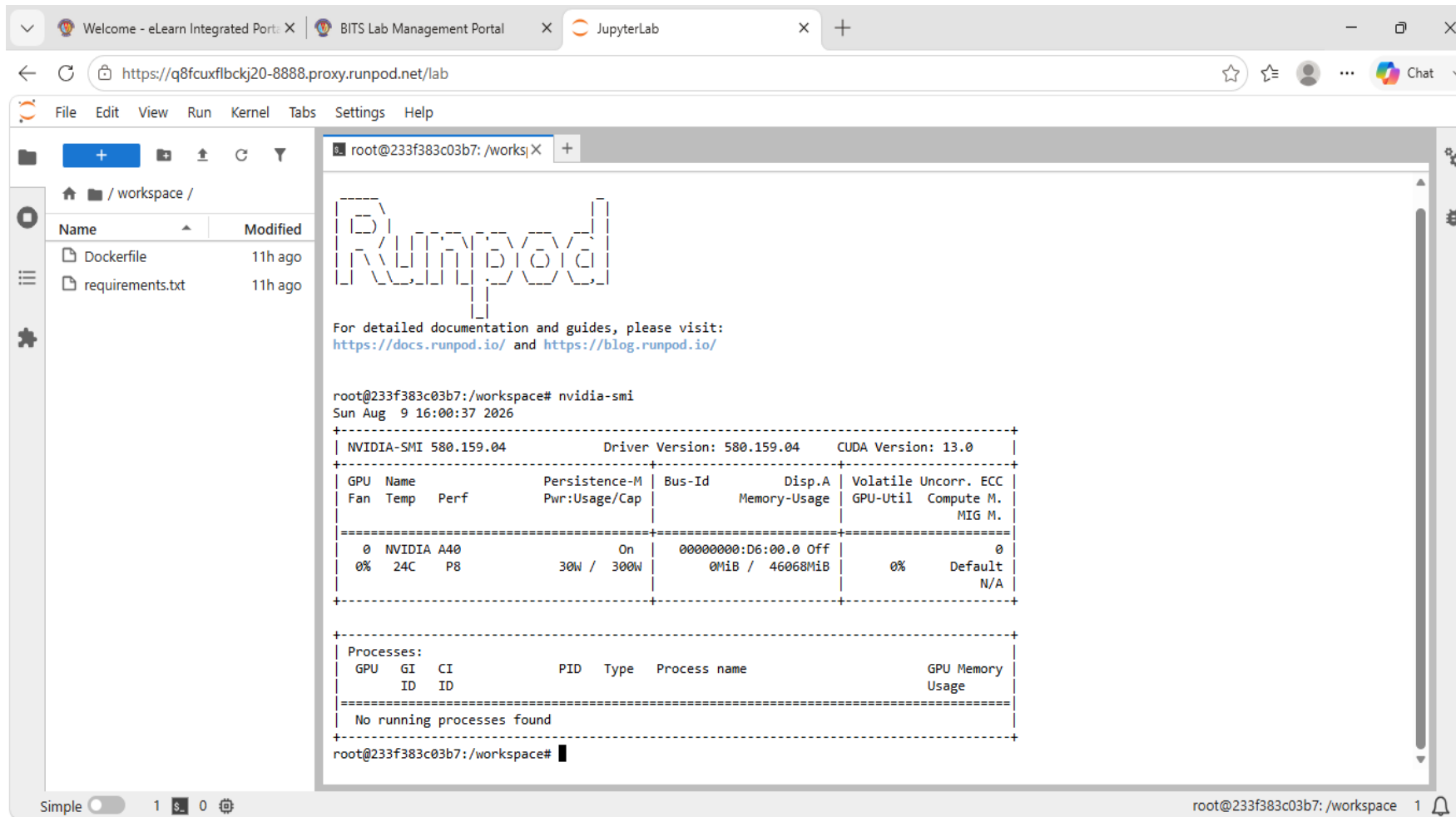
Step 7 - Work in JupyterLab

Use JupyterLab to create notebooks, run Python code, use the terminal, and work with the assigned GPU environment.



Step 8 - Verify GPU Availability

Open a JupyterLab Terminal and run 'nvidia-smi' to verify the assigned GPU, driver version, CUDA version, and current GPU usage.



The screenshot shows a JupyterLab interface with a terminal window open. The terminal displays the output of the 'nvidia-smi' command, which provides detailed information about the GPU hardware, driver version, and current usage.

```
root@233f383c03b7: /workspace# nvidia-smi
Sun Aug 9 16:00:37 2026

+-----+
| NVIDIA-SMI 580.159.04                  Driver Version: 580.159.04      CUDA Version: 13.0     |
+-----+-----+
| GPU   Name           Persistence-M| Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan  Temp  Perf    Pwr:Usage/Cap |      Memory-Usage | GPU-Util  Compute M. |
|-----+-----+
| 0   NVIDIA A40           On       | 00000000:06:00:0 Off |             0        |
| 0%   24C    P8         30W / 300W | 0MiB / 4606MiB | 0%      Default     |
+-----+-----+

+-----+
| Processes:                               |
|  GPU   GI    CI          PID    Type   Process name                      GPU Memory |
|  ID   ID                                   |             Usage   |
+-----+-----+
| No running processes found              |
+-----+

root@233f383c03b7: /workspace#
```

The interface also shows a file explorer on the left with files 'Dockerfile' and 'requirements.txt' in the '/workspace/' directory. The top of the browser window shows the URL 'https://q8fcuxflbkj20-8888.proxy.runpod.net/lab' and several open tabs including 'Welcome - eLearn Integrated Port...', 'BITS Lab Management Portal', and 'JupyterLab'.

Step 9 - Stop the GPU Lab

Save your work and click 'Stop Lab' to end the running GPU session.

The screenshot shows a web browser window with the URL <https://prayogshala.bits-pilani.ac.in/dashboard>. The page displays a 'GPU Lab Session' for 'DUMMY GPU - CSIST112'. The session status is 'RUNNING'. A yellow warning box states: 'Sessions auto-terminate if idle for 15 minutes (no heartbeat)'. The session is associated with 'A40' hardware. The budget status is shown as 'BUDGET REMAINING \$1.93' and 'SPENT \$0.00'. The duration is '0 min'. A red arrow points to a red button labeled 'Stop Lab' with a power icon. Below the button, it shows '1.93 / 2.00 remaining'. A 'Close' button is at the bottom left of the session details panel.

GPU Lab Session
DUMMY GPU - CSIST112

- Sessions auto-terminate if idle for 15 minutes (no heartbeat)

• RUNNING A40

BUDGET REMAINING \$1.93

SPENT \$0.00

DURATION 0 min

Click here

Stop Lab

Credit History 1.93 / 2.00 remaining

Close

Important instructions

- 💰 USD is deducted every minute while the GPU session is running.
- 📁 Save your work before stopping the lab — the pod is destroyed on stop.
- 🎯 GPU availability can vary. The portal displays GPUs as Available or Limited.
- 📊 The session window displays budget remaining, amount spent and session duration.
- 🎯 At times, GPU might be unavailable or become unavailable in the platform during the launch. Please try launching the lab after a few minutes.

Technical SPOC: Mr. Bangaru Santosh Kumar
Mail id: csislabsupport@wilp.bits-pilani.ac.in

Thank You